

VIVEK PATIL

BACKEND & PLATFORM ENGINEER | DISTRIBUTED SYSTEMS, RELIABILITY & OBSERVABILITY

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SUMMARY

Backend engineer with 12+ years of experience designing, delivering, and operating distributed systems in production, including 7+ years at Microsoft on Office for the web. Own prefetching and caching services serving millions of users per hour, and lead a team of 3–4 engineers through design, delivery, and on-call. Proven record improving latency, availability, release safety, and operational simplicity through system design, performance engineering, observability, and incident response. Built an internal retrieval-based AI copilot for production troubleshooting; growing focus on LLM and AI infrastructure. Known for solving difficult technical problems, mentoring engineers, and making pragmatic tradeoffs across performance, correctness, reliability, and delivery speed.

CORE SKILLS

Languages: C#, .NET, Python, Java, TypeScript, JavaScript, C++

Distributed Systems: Backend Services, Service-Oriented Architecture, REST APIs, State and Data Flow, Scalability, High Availability, Failure Handling, System Design

Performance & Reliability: Latency Optimization, Caching, Prefetching, Load and Performance Testing, SLIs/SLOs, Incident Response, Root-Cause Analysis, Graceful Rollouts

Observability: Metrics, Tracing Concepts, At-Scale Logging, Telemetry Pipelines, Dashboards, Alerts, p95/p99 Analysis, Release Health Gates, Azure Data Explorer (Kusto)

Cloud & Infrastructure: Azure, AWS, Docker, Kubernetes, AKS, ECS, EKS, Service Bus, CI/CD, Jenkins, Azure DevOps, GitHub Actions

AI & LLM: Retrieval-Augmented Generation (RAG), Retrieval Workflows, Access-Aware Controls, LLM-Powered Internal Tooling, Diffusion Workflows (ComfyUI, Stable Diffusion, Flux)

Data & Integrations: SQL Server, MongoDB, Feed Processing, Real-Time Alerting, Apache Solr, Data Pipelines

Engineering Leadership: Team Leadership, Architecture Reviews, Code Reviews, Technical Documentation, Mentoring, Onboarding, Cross-Team Delivery

PROFESSIONAL EXPERIENCE

Software Engineer II | Microsoft

Apr 2019 - Present

Office for the web (Word, Excel, PowerPoint, OneNote, Visio). Promoted twice; lead a team of 3–4 engineers.

- Led the architecture, implementation, rollout, and production support of Track Changes for Word Online, coordinating across engineering and product teams and contributing \$10M in annual revenue impact through migration from legacy systems.
- Own prefetching and caching services serving millions of users per hour; optimized backend and client-serving paths with caching, prefetching, and dynamic loading, reducing visually ready time by 50%.
- Architected a secure production-debugging solution using Edge security features so engineers can debug live issues without exposing source maps; adopted across Word, Excel, PowerPoint, OneNote, and Visio, reducing SLA response time by 50% and helping mitigate high-severity security incidents (more than \$20M in avoided impact).
- Defined SLIs/SLOs for critical user journeys and instrumented p95/p99 latency, error-rate telemetry, dashboards, alerts, and release health gates to improve service reliability and rollout confidence.
- Redesigned release practices and CI/CD workflows, reducing release cycles from four weeks to four days through staged and canary deployments, automated validation, rollback playbooks, and operational readiness checks.
- Built a unified performance testing framework that reduced environment setup from 1-2 days to 10 minutes and enabled repeatable validation of critical paths across environments.
- Built an internal AI troubleshooting copilot that combined engineering knowledge sources, retrieval workflows, and access-aware controls to reduce investigation effort for recurring production issues.
- Designed telemetry and data pipelines in Azure Data Explorer to analyze application benchmarks and provide automated dashboards for engineering decisions and production monitoring.
- Owned production issues end to end through diagnosis, mitigation, root-cause analysis, and follow-up reliability improvements; mentored and onboarded engineers and led design and code reviews.

Software Engineer | Sizmek, Inc. (formerly Rocket Fuel)**Jun 2017 - Apr 2019***Ad-serving and data-pipeline engineering for a global ad platform.*

- Refactored ad-serving code to optimize Ad Tag expansion, improving performance by 25% and increasing rendering efficiency for customer-facing traffic.
- Developed real-time monitoring and alerting for a data-processing pipeline, improving issue detection and enabling 30% faster resolution.
- Designed and implemented an automated Ad Ops system that reduced deployment effort by 90% and improved the scalability and repeatability of campaign workflows.
- Integrated feed-processing services with Sizmek MDX to transform source feeds into reliable CSV and XML outputs across backend workflows.
- Implemented JavaScript compression techniques that reduced page load time by 15% and improved end-user performance.

Data Science and Engineering Intern | Scry Analytics, Inc.**Jan 2017 - May 2017**

- Developed Python modules and similarity algorithms for a suggestion system that achieved 90% accuracy in identifying related cases.
- Built an end-to-end MEAN stack application that operationalized analytics outputs into a usable internal workflow.

Software Engineer | Digite Infotech Pvt. Ltd.**Nov 2012 - Jul 2015**

- Refactored application code and optimized database tables, reducing query execution time from one minute to 10 seconds.
- Developed a configurable reporting system that enabled on-demand report generation and reduced turnaround time by 70%.
- Mentored and onboarded engineers by sharing system knowledge, development practices, and troubleshooting guidance.

EDUCATION

Master of Science, Computer Science - University of Southern California, Viterbi School of Engineering, Los Angeles (2017)

Bachelor of Engineering, Computer Engineering - Mumbai University (2012)

SELECTED PROJECTS

AI and Diffusion Workflows

- Built generative image and video workflows using ComfyUI, Stable Diffusion, Flux, and Wan Video, evaluating model chaining, execution speed, output quality, and usability tradeoffs.

Web Crawler and Search

- Developed a crawler with Crawler4j and implemented indexing, ranking, auto-suggest, and spell-correction workflows using Apache Solr.